

SOLANA × MESHTASTIC



SOLTASTIC

Offline transaction relay for Solana

When the mobile internet fails, Soltastic keeps transactions moving.



Resilient payment infrastructure for the decentralized edge.

Connectivity fails at the worst possible moment.

Mobile payments, crypto wallets, and field operations depend on stable internet.
When the network goes down — users lose the ability to transact.



Network Overload

Festivals, emergencies, and mass events saturate mobile towers. Payments stall exactly when they're needed most.



No Coverage

Remote areas, disaster zones, and field operations have weak or zero mobile signal. Transactions are impossible.



Censorship & Disruption

Authoritarian shutdowns and infrastructure attacks cut off financial access for entire populations.

Every failed transaction is a missed opportunity — or a crisis.

Five forces are converging to make this the right moment.

01 Mobile Crypto Growth

Wallet adoption is accelerating globally. Solana leads in speed and cost, making it ideal for everyday payments.

02 LoRa Devices Cheapening

Meshtastic LoRa hardware costs under \$30 and is growing a global community of mesh operators.

03 Durable Nonce is Live

Solana's durable nonce feature makes delayed or offline transaction submission technically viable — no workarounds.

04 DePIN Infrastructure Boom

Decentralized physical infrastructure is a major emerging sector. Investors and builders are actively seeking new DePIN primitives.

05 Demand for Resilience

Governments, NGOs, and enterprises are investing in censorship-resistant and disaster-ready financial infrastructure.

THE SOLUTION

Sign offline. Deliver through mesh. Transact on Solana.



Users sign Solana transactions locally inside their existing wallet (Phantom, Solflare). Private keys never leave the device.



Compact transaction data is broadcast through a Meshtastic LoRa mesh network — no mobile internet required.



An online relay server (M-server) receives the mesh packet, verifies, reconstructs, and submits the transaction to Solana.



The confirmed transaction hash travels back through the mesh to the user. The whole flow is end-to-end verifiable.



USER WALLET

Signs locally



SOLTASTIC APP

Packages transaction



MESH NETWORK

LoRa multi-hop



M-SERVER → SOLANA

Submit & confirm

No trusted intermediary. No custodial risk. Fully verifiable.

A full offline-to-onchain transaction pipeline.



← Confirmation hash returns through mesh to user

Security guarantee:

Private keys never leave the device. The transaction is signed before entering the mesh. The M-server only submits a pre-signed transaction it cannot modify. Even a compromised relay cannot steal funds.

Purpose-built components for offline transaction continuity.



Solana Durable Nonce

Transactions remain valid for extended periods. Delayed or offline delivery is possible without expiry failures.



Local Wallet Signing

Phantom & Solflare sign locally. Soltastic never touches private keys — signing is fully wallet-native.



Meshtastic LoRa Mesh

Sub-GHz radio mesh with multi-hop delivery. Operates without any internet infrastructure across km-scale distances.



M-Server Relay

Internet-connected relay nodes reconstruct, verify, and submit transactions to Solana RPC endpoints.



Compact Payload Protocol

Transaction metadata + signature encoded for low-bandwidth LoRa transmission. Minimal overhead, maximum compatibility.



Server-Side Verification

M-server verifies signature integrity before submission. Invalid or tampered transactions are rejected.

Wherever connectivity fails, Soltastic fills the gap.



Festivals & Events

Overloaded towers make mobile payments fail at peak moments. Soltastic keeps vendor and ticket transactions alive.



Remote Areas

Expeditions, agriculture, mining, and energy field teams can transact without cellular coverage.



Disaster Response

Emergency responders and NGOs need financial coordination when infrastructure is down.



Censorship-Resistant

Users in countries with internet shutdowns can still access decentralized financial services.



Solana & Crypto Native

Power users, DeFi traders, and dApp operators who need guaranteed transaction delivery.



Meshtastic Operators

The global Meshtastic community becomes the first infrastructure operator base — incentivized relay runners.

Multiple revenue channels across users, operators, and enterprises.

01 Per-Transaction Relay Fees

Micro-fees charged on each transaction routed through the M-server relay network. Scalable and automatic.

02 Gateway Kits

Bundled hardware + software for events, NGOs, and field teams needing turn-key offline payment infrastructure.

03 SDK & API Integrations

Developer-facing SDK for wallet providers and dApps to integrate Soltastic's offline transaction protocol natively.

04 Operator Relay Network

Community M-server operators earn a share of relay fees. Soltastic earns a network protocol fee on each relay.

05 Enterprise Deployments

Resilient payment infrastructure licensing for governments, utilities, mining companies, and critical operations.

The moat is not LoRa. It's the full protocol stack.



Offline Solana transaction protocol with durable nonce flow



Wallet-native UX — users sign in their existing wallet, zero custodial risk



Mesh-aware retry and delivery logic ensures transactions are not lost



Relay reputation and payment logic — a permissioned operator economy



Transport-agnostic architecture: BLE, SMS, satellite, Wi-Fi Direct ready



First-mover integration with the global Meshtastic community operator base

Capability Matrix

	Solastic	SMS Pay	Sat TX
Offline sign	✓	⚠	✓
Low cost	✓	✓	✗
No custodian	✓	✗	⚠
Mesh delivery	✓	✗	✗
Solana native	✓	✗	✗
Open protocol	✓	✗	✗

Working prototype on Solana devnet. Proof of concept validated.

Prototype Capabilities

- ✓ Connect to Meshtastic device over Bluetooth
- ✓ Request wallet balance and durable nonce from M-server via mesh
- ✓ Sign transaction locally inside Phantom / Solflare wallet
- ✓ Send transaction metadata + signature through Meshtastic mesh
- ✓ M-server reconstructs, verifies, and submits to Solana RPC
- ✓ Confirmed transaction hash returned to user through mesh

Performance Metrics

Transaction delivery rate	To be measured
Median confirmation time	To be measured
Max tested mesh distance	To be measured
Payload size / transaction	To be measured
Relay cost / transaction	To be measured
Meshtastic nodes tested	To be measured
Wallets tested	Phantom, Solflare

A community-first expansion strategy.

1

LAUNCH

Now → 6 months

- Win Solana hackathons & ecosystem grants
- Open-source the core protocol to build community
- Deploy pilots with Meshtastic community operators
- Demos at crypto events and DePIN conferences

2

GROW

6 → 18 months

- Wallet SDK integrations with Phantom & Solflare
- Event payment gateway kits for festivals & markets
- Operator relay network with fee-sharing model
- Field pilots: NGOs, remote enterprise, agriculture

3

SCALE

18 months+

- Multi-transport support: BLE, SMS, satellite
- Enterprise resilience contracts
- Multi-chain protocol expansion beyond Solana
- Become the standard offline transaction layer

VISION & CLOSING

TODAY

Solana transactions over Meshtastic LoRa — last-mile delivery for the decentralized edge.

TOMORROW

Offline transaction continuity across multiple low-bandwidth transports — LoRa, BLE, SMS, satellite, and beyond.

Soltastic becomes the resilient transaction layer for the offline edge.

*Soltastic keeps digital payments alive
when mobile networks fail.*



Built on Solana • Powered by Meshtastic • Designed for the real world